ModelIt!

Teacher's Guide

Why Is Groundwater Disappearing?

How Human Activity and Natural Cycles Determine the Future of Underground Water

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-ESS3-1, MS-ESS2-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-1, MS-ESS2-4

Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes. Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how agricultural extraction and natural recharge interact to determine aquifer levels, land subsidence, and long-term water security.
Disciplinary Core Idea	**ESS3.A Natural Resources** — Humans depend on Earth's land, ocean, atmosphere, and biosphere for many different resources, including groundwater. The distribution and sustainability of these resources is determined by geoscience processes.
Crosscutting Concept	**Stability and Change** — Students analyze how groundwater systems can remain stable (when recharge equals extraction) or change dramatically (when extraction exceeds recharge), and identify the tipping points where change becomes irreversible.

CAST Testing Alignment

MULTIPLE CHOICE:

A farming region pumps 500 million gallons per year from its aquifer. Rainfall recharges the aquifer at 200 million gallons per year. At this rate, which of the following will occur, and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why land subsidence is considered irreversible and how it creates a feedback loop that worsens groundwater depletion. Include the relationship between aquifer level and storage capacity in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Groundwater — water stored in underground rock formations called aquifers — supplies drinking water for nearly half the world's population and irrigates much of the world's farmland. Yet this invisible resource is being depleted at alarming rates, creating a slow-motion crisis that threatens food production and water security worldwide.

What Is an Aquifer?

An aquifer is an underground layer of permeable rock, sand, or gravel that holds and transmits water. Water fills the tiny spaces (pores) between rock particles, like water in a sponge. Major aquifers can be enormous — the Ogallala Aquifer underlies 174,000 square miles across eight US states. Some aquifers were filled millions of years ago and receive very little new water today (fossil aquifers).

The Recharge Problem

Rainfall recharge is the process by which surface water percolates through soil and rock to reach an aquifer. This process is slow — water may take years to decades to travel from the surface to a deep aquifer. In arid regions, very little rain reaches the aquifer because most evaporates or is consumed by plants. When pumping exceeds recharge, the aquifer is essentially being mined like a non-renewable resource.

Land Subsidence: The Irreversible Consequence

When water is pumped from an aquifer, the pore spaces that held the water can collapse and compact under the weight of overlying land. This causes the ground surface to sink — a process called subsidence. In California's Central Valley, some areas have sunk over 28 feet since the 1920s. Critically, compacted rock cannot re-expand — even if the aquifer is refilled, it can never hold as much water as before. Subsidence is permanent.

The Global Groundwater Crisis

The world's largest aquifers are being depleted. The Ogallala Aquifer (US Great Plains) has lost an average of 15 feet of water since 1950, with some areas losing over 150 feet. India's aquifers support 60% of irrigated agriculture but are declining rapidly. The North China Plain aquifer drops 3 feet per year. NASA's GRACE satellites measure groundwater from space and show that 21 of Earth's 37 major aquifers are being drained faster than they're refilled.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify agricultural irrigation, rainfall recharge, aquifer level, land subsidence, and long-term water security as system components.

E **Establish Relationships**
Students determine that irrigation depletes aquifer levels while rainfall recharges them, dropping aquifer levels cause land subsidence, and the balance determines long-term water security.

V **Visualize & Model**
Students build the groundwater model showing how extraction and recharge rates determine aquifer sustainability and the cascading effects of depletion.

E **Evaluate Outcomes**
Students compare balanced vs. over-extraction scenarios to verify that sustainable pumping requires extraction at or below recharge rates, and observe irreversible subsidence effects.

R **Revise & Extend**
Students add urban water demand, contamination risk, or variable recharge rates to model more complex groundwater dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Beneath your feet right now, there might be an ocean of water. And it's disappearing."*

Watch For

Show aquifer cross-section. Ask: Where does well water actually come from?

Slide 2: Learning Objectives

3 min

Say: *"Today we model why underground water that took millennia to accumulate is vanishing in decades."*

Watch For

Pre-teach aquifer (underground reservoir), recharge (natural refilling), subsidence (ground sinking).

Slide 3: Big Question

3 min

Say: *"The Ogallala Aquifer feeds America's breadbasket. Some areas have lost half their water in 70 years. Why?"*

Watch For

Show before/after aquifer level maps. Ask: What happens when you drain a sponge faster than it can soak up water?

Slide 4: LEVER Framework

2 min

Say: *"We'll model the balance between what goes in (rain) and what comes out (pumping)."*

Watch For

Preview that this is fundamentally a balance problem — input vs. output over time.

Slide 5: Activity 1: Components

8 min

Say: *"Irrigation and rainfall are our inputs. Aquifer level, subsidence, and water security are the outputs."*

Watch For

Sort components. Discuss why irrigation and rainfall are external (human choice and weather).

Slide 6: Activity 2: Connections

8 min

Say: *"Pumping drains the aquifer. Rain refills it. When the aquifer drops, the ground sinks — permanently."*

Watch For

Map relationships. Highlight the irreversibility of subsidence — this is the critical insight.

Slide 7: Activity 3: Simulation

10 min

Say: *"Compare sustainable pumping vs. over-extraction over 50 years. The difference is dramatic."*

Watch For

Students predict before running. Focus on the long-term divergence and the subsidence feedback loop.

Slide 8: Discoveries

5 min

Say: *"This isn't a future problem — it's happening now. Twenty-one of Earth's major aquifers are being drained."*

Watch For

Show NASA GRACE satellite data. Connect to local water sources and regional agriculture.

Slide 9: STEM Challenge

12 min

Say: *"A farming community needs your help. Design a plan that saves both the farms and the aquifer."*

Watch For

Teams create management plans using model evidence. Present trade-offs and justify recommendations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Agricultural Irrigation, Rainfall Recharge
- **INTERNAL:** Aquifer Level, Land Subsidence, Long-term Water Security

Relationships

- **Agricultural Irrigation to Aquifer Level:** NEGATIVE (-) — More pumping for irrigation removes water from the aquifer, lowering the water table.
- **Rainfall Recharge to Aquifer Level:** POSITIVE (+) — More rainfall percolation adds water to the aquifer, raising or stabilizing the water table.
- **Aquifer Level to Land Subsidence:** NEGATIVE (-) — Lower aquifer levels cause underground pore spaces to collapse, increasing land subsidence at the surface.
- **Land Subsidence to Aquifer Level:** NEGATIVE (-) — Subsidence permanently compacts aquifer rock, reducing its storage capacity — even if water is added, the aquifer holds less than before.
- **Aquifer Level to Long-term Water Security:** POSITIVE (+) — Higher aquifer levels mean more water available for future use, increasing long-term water security for the region.

Key Discoveries

- When extraction exceeds recharge, aquifer levels decline steadily — this is called groundwater mining
- Rainfall recharge is extremely slow — water may take decades or centuries to percolate down to deep aquifers
- As aquifer levels drop, the empty pore spaces in rock and sediment compact under the weight of overlying land, causing permanent land subsidence
- Land subsidence is irreversible — once the ground sinks, the aquifer can never hold as much water again even if refilled
- Long-term water security requires that extraction rates stay at or below natural recharge rates

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Groundwater is an underground lake or river"	Groundwater doesn't exist in open underground caverns (except in rare karst geology). It fills tiny pore spaces between rock and sediment particles, like water in a sponge. An aquifer is rock or sediment whose pores are saturated with water — it looks like wet sand, not an underground pool.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Rain immediately refills the aquifer"	Rainfall recharge is extremely slow. Water must percolate through soil, clay layers, and rock — a journey that can take months, years, or even centuries to reach deep aquifers. Most rainfall evaporates, runs off into rivers, or is absorbed by plant roots long before reaching the aquifer.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can always drill deeper to find more water"	Drilling deeper wells is a temporary solution that accelerates the problem. Deeper pumping lowers the water table faster, increases energy costs, and can access lower-quality water. It also increases subsidence by pulling water from deeper rock that is even more prone to compaction. Eventually, there is no deeper water to reach.